Rational Number Project

Level 2 / Lesson 8 / Overview

Students review how to model fractions with fraction circles by exmining fractions close to 1, 0, and 1/2. They order two fractions by comparing to 1/2.

Materials

- Fraction Circles for students and teacher
- Rods, chips, & paper for folding (for teacher)
- Student Pages A, B

Teaching Actions

1. Ask students to name the different manipulative materials they have used to show fractions.

2. Have students share their preferences, explaining why they may like one model over another.

3. At the overhead show the fraction 2/3 using the rods. Ask students to:

   (a) Show the same fraction with fraction circles.

   (b) Explain how the two models are alike and how they are different.

   (c) Say the name for the fraction.

   (d) Explain why it has that name.

   (e) Write the symbol.

   (f) Show the fraction with the circles but using a different unit.

4. Repeat showing a fraction with the chips or by paperfolding. Students model with the fraction circles.

Comments

1. Lesson 8 is the first of two lessons that uses fraction circles. The lessons lead students to develop four informal ordering strategies [see the teacher's guide for detailed discussion on these strategies]. Students use their mental images of fraction circles to order fractions. Much of the ground work for developing the mental images that are part of students "number sense" has been done in Level 1. These lessons make the ordering ideas explicit.

Teaching Actions

5. Have students show the following fractions with their fraction circles:

$\frac{11}{12}$ $\frac{3}{4}$ $\frac{8}{9}$ $\frac{14}{15}$ $\frac{7}{8}$

6. Ask students how these fractions are:

   Alike [close to 1; all 1 piece away from whole circle.]

   Different [Not the same size; the smaller the piece size, the larger the fraction.]

7. Ask students to rewrite the list from smallest to largest.

8. Have students show the following fractions with their fraction circles:

$\frac{1}{9}$ $\frac{2}{15}$ $\frac{2}{12}$ $\frac{1}{4}$ $\frac{1}{8}$

9. Ask students how the above fractions are alike and different. Have them order the fractions from smallest to largest. [Verify answer with the fraction circles; use the whole circle as the unit].

Comments

2. You are trying to help students think of fractions that are close to 1. They will realize that if the numerator and denominator of a fraction are close to each other, the size of the fraction is close to 1.

3. Here students look at fractions that are small. They should realize that to show a fraction close to zero, increase the denominator and have the numerator be close to one.

Teaching Actions

10. Present this problem to students to do together in groups.

    Using circles find 5 examples of fractions that are equal to 1/2, 5 examples that are a little bigger than 1/2, and 5 examples that are a little smaller than 1/2. Record the fractions on a piece of paper.

    Fractions just over 1/2: _____

    Fractions equal to 1/2: _____

    Fractions just under 1/2: _____

11. Have students share their results in a large group. Ask them if they see anything special about the numerator and denominator of fractions greater than 1/2; less than 1/2; equal to 1/2. Have them apply their thinking to this example:Is 5/8 >, < or = 1/2?

    - 5/8 > 1/2 because:
    - "If I double the numerator and the answer is greater than the denominator, then the fraction is greater than 1/2."
    - "Numerator is more than half the denominator."
    - "4/8 = 1/2 so 5/8 is more."

12. Conclude by asking children to:

    (a) Imagine a fraction close to one. Describe and name it. [Circle divided into 50 parts, 49/50 would be almost the whole area.]

    (b) Imagine a fraction close to zero. Describe and name it.

    (c) Imagine a fraction a bit more than 1/2. Describe and name it.

    (d) Imagine a fraction a bit less than 1/2. Describe and name it.

13. Student Pages A-B reinforce the ordering of fractions.

Comments

4. Students consider 1/2 as a reference point.

5. Now you start making symbolic representations explicit for fractions close to 1, zero, and1/2 through the use of mental images.

Fraction Estimation

Picture these fractions in your mind. Is the fraction greater than 1/2 or less than 1/2? Put > or < signs in each box to show your answer.

$\frac{2}{3}$ ☐ $\frac{1}{2}$ $\frac{1}{4}$ ☐ $\frac{1}{2}$ $\frac{15}{18}$ ☐ $\frac{1}{2}$

$\frac{3}{10}$ ☐ $\frac{1}{2}$ $\frac{3}{9}$ ☐ $\frac{1}{2}$ $\frac{6}{10}$ ☐ $\frac{1}{2}$

$\frac{6}{11}$ ☐ $\frac{1}{2}$ $\frac{5}{12}$ ☐ $\frac{1}{2}$ $\frac{4}{6}$ ☐ $\frac{1}{2}$

$\frac{5}{12}$ ☐ $\frac{1}{2}$ $\frac{3}{7}$ ☐ $\frac{1}{2}$ $\frac{9}{20}$ ☐ $\frac{1}{2}$

Complete these fractions so that they are all close to 1/2, but just a bit bigger.

$\frac{\quad}{10}$ $\frac{\quad}{12}$ $\frac{\quad}{11}$ $\frac{\quad}{20}$ $\frac{\quad}{13}$ $\frac{4}{\quad}$ $\frac{7}{\quad}$

Fraction Estimation

Picture these fractions in your mind. Order the fractions in each set from smallest to largest.

$\frac{9}{16}$	$\frac{8}{9}$	$\frac{1}{14}$
___	___	___

$\frac{19}{20}$	$\frac{2}{22}$	$\frac{6}{14}$
___	___	___

$\frac{99}{100}$	$\frac{6}{11}$	$\frac{3}{100}$
___	___	___

$\frac{6}{7}$	$\frac{2}{88}$	$\frac{6}{10}$
___	___	___

$\frac{10}{13}$	$\frac{2}{47}$	$\frac{4}{7}$
___	___	___

$\frac{3}{50}$	$\frac{13}{15}$	$\frac{5}{11}$
___	___	___

What fractions are close to 0? _____

What fractions are about 1/2? _____

What fractions are close to 1? _____

On the back of the sheet describe your strategies for ordering each set of fractions.